

GEOBIOLOGY



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UCD: Geobiology (Sumner)

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TABLE OF CONTENTS

Licensing

1: Life and Earth

- 1.1: Daisyworld
- 1.2 Interactions Among Organisms

2: What is life?

- 1.1: What is life?

3: Genes to Enzymes

- DNA, RNA, Genes and Chromosomes
- DNA Replication
- How Asexual Prokaryotes Achieve Genetic Diversity
- Mutations
- Operon Theory
- Protein Synthesis (Translation)
- RNA Transcription
- What Are Genes?

4: Tracing the History of Evolution

- 4.1: Phylogenetic Trees
- 4.2: Limitations of Phylogenetic Trees
- 4.3: The Levels of Classification

5: Structure of life

- 3.1: Unique Characteristics of Prokaryotic Cells
- 3.2: Unique Characteristics of Eukaryotic Cells
- 3.3: Viruses
 - 3.3A: Viruses
 - 3.3B: The Viral Life Cycle

6: Microbial Metabolism

- 8.1: Energy, Matter, and Enzymes
- 8.2: Catabolism of Carbohydrates
- 8.3: Cellular Respiration
- 8.4: Fermentation
- 8.5: Catabolism of Lipids and Proteins
- 8.6: Photosynthesis and the Importance of Light
- 8.7: Biogeochemical Cycles

7: How Microbes Grow

- 5.1: How Microbes Grow

[Index](#)

[Glossary](#)

[Detailed Licensing](#)

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

Front Matter

[TitlePage](#)

[InfoPage](#)

[Table of Contents](#)

[Licensing](#)

UCD: Geobiology (Sumner)

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TABLE OF CONTENTS

Licensing

1: Life and Earth

- 1.1: Daisyworld
- 1.2 Interactions Among Organisms

2: What is life?

- 1.1: What is life?

3: Genes to Enzymes

- DNA, RNA, Genes and Chromosomes
- DNA Replication
- How Asexual Prokaryotes Achieve Genetic Diversity
- Mutations
- Operon Theory
- Protein Synthesis (Translation)
- RNA Transcription
- What Are Genes?

4: Tracing the History of Evolution

- 4.1: Phylogenetic Trees
- 4.2: Limitations of Phylogenetic Trees
- 4.3: The Levels of Classification

5: Structure of life

- 3.1: Unique Characteristics of Prokaryotic Cells
- 3.2: Unique Characteristics of Eukaryotic Cells
- 3.3: Viruses
 - 3.3A: Viruses
 - 3.3B: The Viral Life Cycle

6: Microbial Metabolism

- 8.1: Energy, Matter, and Enzymes
- 8.2: Catabolism of Carbohydrates
- 8.3: Cellular Respiration
- 8.4: Fermentation
- 8.5: Catabolism of Lipids and Proteins
- 8.6: Photosynthesis and the Importance of Light
- 8.7: Biogeochemical Cycles

7: How Microbes Grow

- 5.1: How Microbes Grow

[Index](#)

[Glossary](#)

[Detailed Licensing](#)

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Life and Earth

Earth is the only known inhabited planet, and the geological record demonstrates that life and Earth have co-evolved for billions of years in a closely linked system. Life is sustained by energy and material exchanges among Earth's interior, Earth's surface, and the Sun. In turn, life has changed the surface chemistry of Earth. These changes have propagated into Earth's interior through subduction. Through time, the mantle, lithosphere, hydrosphere, atmosphere, and biosphere have slowly evolved, with intervals of rapid change due to historical events (e.g. impacts, evolution of land plants, etc.). Both incremental evolution and historical events are reflected in the co-evolution of Earth and life. This evolution is reflected in biogeochemical cycles that extend from the mantle to the top of the atmosphere, with ecosystems evolving in response to perturbations and inducing planetary-scale changes in Earth's surface processes.

The field of Geobiology focuses on understanding these changes using diverse fields of study. Geobiologists can study modern life through ecology, organismal processes, genetics, and biogeochemistry. They can study the history of these interactions through geological studies, paleontology, geochemical approaches, and modeling. They can study these processes on other planets by applying what we understand of interactions on Earth to our observations from missions and telescopes. The key aspect that makes a study part of Geobiology is that it provides insights into how life and Earth (or other planets or moons) interact through time.

Sometimes it helps to look at a simplified system to understand a more complex one. The first part of this chapter introduces a very simple biosphere on a very simple world: Daisyworld (Watson and Lovelock, 1983). Black and white daisies on a planet receive energy from a star and influence the planet's temperature. We ignore all the biological aspects of daisies except their preferred temperature for sprouting and growth, plus their need for somewhere to grow. Even though Daisyworld represents a very simple ecosystem, it is still an interesting planet!

The second part of this chapter discusses some of the interactions among organisms that create complexity in ecosystems. In all cases, these interactions occur in the context of our planet and the resources it provides. While reading about interactions types, think about the interactions among the daisies in Daisyworld. How many different types of interactions are there? Can they be classified into the interaction types described in the second part of the chapter? If this was a real ecosystem, what other organisms would be needed to allow the daisies to grow? How would any of these needs affect the simple Daisyworld model?

Thumbnail: An artist's impression of ice age Earth at glacial maximum. CC-BY-SA 3.0 Unported; Ittiz via [wikipedia](#)).

[1.1: Daisyworld](#)

[1.2 Interactions Among Organisms](#)

References

Watson, Andrew J. & James E. Lovelock (1983) Biological homeostasis of the global environment: the parable of Daisyworld, *Tellus B: Chemical and Physical Meteorology*, 35:4, 284-289, DOI: [10.3402/tellusb.v35i4.14616](https://doi.org/10.3402/tellusb.v35i4.14616)

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1.1: Daisyworld

(Content based on: Andrew J. Watson & James E. Lovelock (1983) Biological homeostasis of the global environment: the parable of Daisyworld, *Tellus B: Chemical and Physical Meteorology*, 35:4, 284-289, DOI: [10.3402/tellusb.v35i4.14616](https://doi.org/10.3402/tellusb.v35i4.14616))

Summary

Life and Earth have interacted over billions of years, with life profoundly changing environments on Earth. In turn, Earth provides the boundary conditions for life. Lovelock (1983) and Watson and Lovelock (1983) proposed that life and Earth be considered as two parts of a coupled system. This system is too complex to fully model, but some of the behaviors of the interactions can be explored with a very simple system proposed by Watson and Lovelock (1983): Daisyworld. Daisyworld is a biosphere consisting of two species of daisy with different colors. “The growth rate of the daisies depends on only one environmental variable, temperature, which the daisies in turn modify because they absorb different amounts of radiation.” These relationships produce feedbacks that can stabilize or destabilize the temperature, depending on the temperature growth preferences for the two daisy species. Temperature is stabilized when the growth-temperature curve is peaked with darker (more heat absorbing) daisies growing at lower temperatures than white (more heat reflecting) daisies. Watson and Lovelock (1983) propose that these sorts of feedbacks stabilize temperature on Earth.

Introduction

Environments on Earth are modified by living things at scales from individual chemicals to the oxidation state of Earth’s surface. Even the formation of granite has been proposed as due to biological carbon cycling (e.g. ref). “In turn, geophysical and geochemical constraints have shaped the evolution of life and continue to dictate what type of life, and how much of it, can colonize” environments on Earth (Watson and Lovelock, 1983). Watson and Lovelock (1983) created a coupled system with two components: life and Earth. In their model, perturbations of life affect Earth and of Earth reflect life, leading to feedbacks between the two components. The coupling between the components may amplify a perturbation in a positive feedback or dampen it in a negative feedback.

Studying these feedbacks is difficult as demonstrated by the challenges of modeling climate change (e.g. refs). Non-linear feedbacks produce chaotic systems; for example the dynamics of the atmosphere led to the Lorenz Equations, which are a classic example of systematic behaviors but unpredictable results in detail. Consider a model that includes all of life and Earth processes over geological time; it is too complex to model in full. However, Watson and Lovelock (1983) proposed “an artificial world, having a very simple biota which is specifically designed to display the characteristic in which [they] are interested—namely, close-coupling of the biota and the global environment.” They proposed a few equations to capture this world and illustrate the importance of life-Earth interactions. The equations can be found in the original reference and are not reproduced here (yet) (Watson and Lovelock, 1983).

Daisyworld Model

Daisyworld is a planet colonized only by two species of daisy of different colors. One species is dark (called “black” here), and one species is light (called “white” here). Black daisies absorb more heat than white daisies do, so the parts of Earth covered by black daisies tends to warm up through time, whereas the parts covered by white daisies tend to cool down relative to bare ground. Watson and Lovelock (1983) propose a set of coupled differential equations to rigorously model the changes in temperature and to calculate the proportion of the planet covered with bare ground and each type of daisy. This model is fun to play with. However, it is also more complicated than we need to understand the feedbacks. Even a system as simple as the GEL 53 Geobiology [Daisyworld Game](#) produces interesting dynamics.

The key to understanding the feedbacks is to explore how the rules of the system either dampen or enhance perturbations. If black daisies grow better at lower temperatures than white daisies, cooler temperatures will lead to more black daisies, which will in turn increase the amount of heat absorbed. The absorption of more heat leads to warmer temperatures. As temperatures increase, more white daisies will grow, leading to less heat absorption and cooler temperatures. In this scenario, perturbations in temperature are dampened by the growth of the daisies, which is a negative (dampening) feedback. In the full Watson and Lovelock (1983) model:

The “environmental feedback” in the model as described so far is strongly limiting on both species of daisy. Black daisies are warmer than white, and tend therefore to be favoured by cooler mean temperatures, yet an increase in the numbers of black daisies

tends to warm the planet. The same goes in reverse for the white daisies. Under these circumstances it is perhaps not surprising that the system exhibits a stable point around which the daisies can successfully homeostat the temperature over a wide range of luminosities.

In contrast, if black daisies grow better at higher temperatures than white daisies, an increase in temperature will lead to more black daisies, which absorb more heat, which increases temperature even more. In this scenario, perturbations in temperature are amplified by the growth of the daisies, which is a positive (amplifying) feedback. Watson and Lovelock (1983) show how this positive feedback can be countered by adding another negative (dampening) feedback process to the system:

Owing to a subtle change of climate, clouds appear on daisyworld. The clouds are light in colour. We will assume that the clouds form only over stands of black daisies because of the rising air generated over these warm spots. Now, therefore, black daisies no longer tend to increase the temperature. Instead, more black daisies mean more white clouds and a colder planet.

However, this change in dynamics has a huge impact on white daisies. They end up going extinct in the Watson and Lovelock (1983) model:

White daisies were not deliberately suppressed, but became extinct of their own accord. White daisies fail because, in the battle for survival of the fittest, they are now distinctly less fit than the black daisies. An increase in either species now tends to cool the planet, but black daisies, being warmer than the mean, thrive better at low temperatures.

The conditions for daisy growth in their model are slightly different than that in the Daisyworld Game; it is not clear to me that white daisies can go extinct in our game without significant rule changes.

Relevance to Earth

The behaviors in Daisyworld are relevant to Earth even though the system is exceptionally simplified. Nearly a century of work developing numerical climate models has demonstrated that feedbacks among the biosphere, hydrosphere, and atmosphere are complexly interrelated (see [Pinho, 2020](#) for a brief explanation). In particular, life has a substantial influence on Earth's temperature through both the heat absorbed and the cycling of greenhouse gases. It is very interesting to read Watson and Lovelock's original section on the relevance of Daisyworld to Earth in the context of our relatively poor understanding of greenhouse processes at that time. Here it is in its entirety:

5. Relevance to the earth

Extrapolation from daisyworld to the earth is, to say the least, rather tenuous at this stage. However, a peaked growth versus temperature curve is a universal property of living things. Furthermore, the biota may have a substantial influence on the earth's temperature via the abundance of green- house gases in the atmosphere. Recently, Owen et al. (1979) and Walker et al. (1981) have speculated that the abundance of atmospheric CO₂ may have been dominant in determining the mean temper- ature of the earth through geological time, in which case the biota as a whole would appear to be depressing the mean temperature below that of the "sterile earth" by tending to reduce atmospheric CO₂ pressures (Lovelock and Watson, 1982). In the present context, neither the direction nor the mechanism whereby life affects the temperature are of themselves important---only the assumption that the biota influence the temperature is required. For the sake of illustration, however, let us suppose that the net effect of life on Earth is to reduce atmospheric carbon dioxide, and that the biota are temperature limited. Thus a decrease

in temperature would lead to an extension of the barren polar regions and would decrease the average level of biological activity over the earth as a whole, while a temperature increase would have the opposite effect. But a decrease in biological activity as a whole would presumably also decrease those activities which tend to reduce atmospheric CO₂. Thus carbon dioxide would increase to oppose the original change. We then have the rudiments of a temperature stabilization system for the earth analogous to that on daisyworld. We can speculate that some such mechanisms may have played a part in regulating the temperature and other environmental variables over the long history of the earth.

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1.2 Interactions Among Organisms

Introduction

In the previous section on Daisyworld, the interactions among daisies appear very simple. Yet Daisyworld has some very interesting behaviors due to feedbacks among black and white daisies and temperature. In real ecosystems, the interactions are more varied and complex. These interactions commonly produce feedbacks in the respective populations of various species. Competition for resources and predation can limit populations of organisms. Mutually beneficial relationships can increase the populations of both species interacting. And the organisms influence their environment, shaping Earth's surface and the resources available to ecosystems. Thus, interactions among species are a major component of how life on Earth persists and evolves through time.

Some Interaction Types

Most interactions between species have produce harms and benefits for the organisms involved, but sometimes the results are closer to neutral. There are typically 6 types of interactions when considering the harms and benefits to each species (Figure), but there are also other ways to frame interactions (see this [Khan Academy video](#) for example).

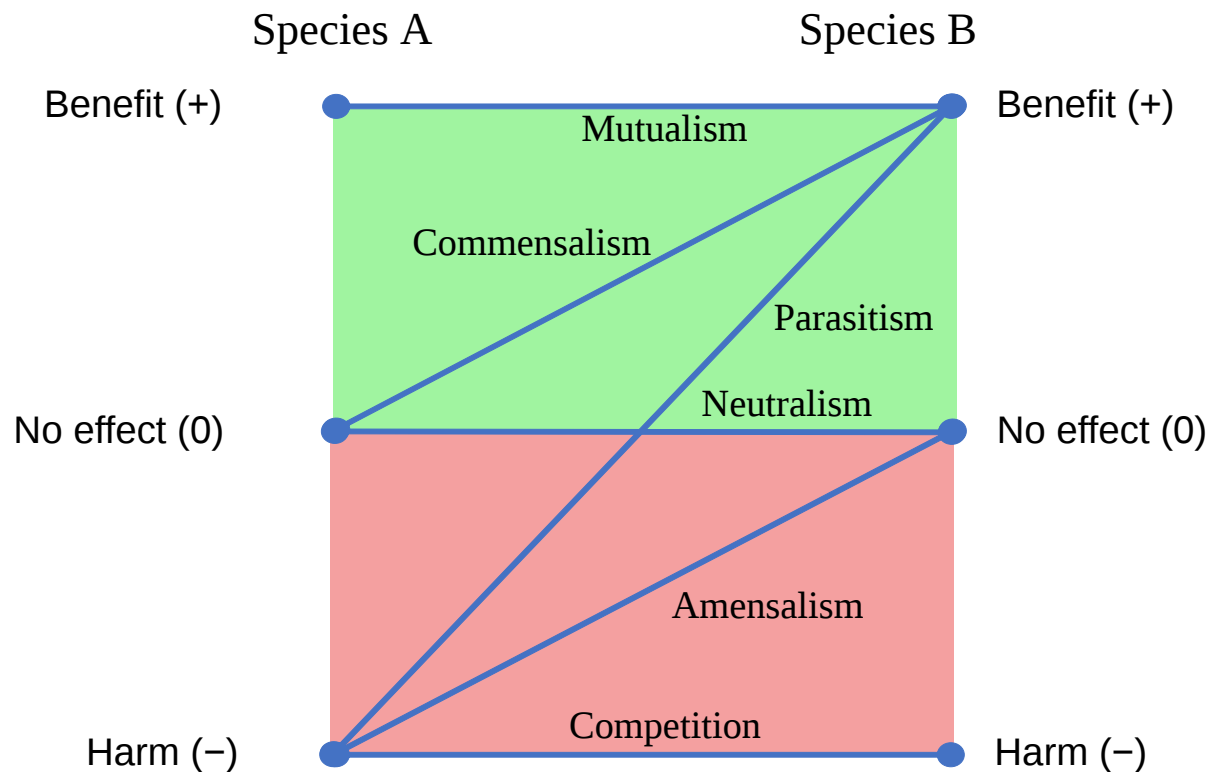


Figure: The benefits and harms to organisms with different types of interactions. Predation has the equivalent effects as parasitism in this diagram. (Diagram by Ian Alexander, Wikipedia, CC BY-SA 4.0)

The following sections include these interactions types, with examples. For the first three interactions types, mutualism, neutralism, and competition, the effects of the interaction are similar on both of the species. For the next three, the effects are asymmetric, with one species benefiting more than the other from the interaction. Note that these categories are neither the only interaction types nor are they able to capture the intricacies of actual interactions among organisms. For example, how does an [opossum pushing a skunk into the water](#) fit within this classification? (Confession: I just wanted to include a link to the video.)

Mutualism

Many interactions result in benefits to both organisms, and for a relationship to be mutually beneficial, both species need to do better when interacting with the other. These relationships include organisms providing resources and protection for each other. Ecosystems are commonly built around these relationships, and mutualistic relationships allow large amounts of biomass to accumulate, say in a rainforest. Organisms helping each other increases the amount and diversity of life that can be supported.

Examples: Insects pollinate flowers, which provide the insects with nectar and pollen as food. Fungi obtain their energy from breaking down dead wood to release nutrients for the living trees. Bacteria in our guts help us digest our food, and we provide them with a stable environment with plenty of food for themselves. The examples are limitless.

Neutralism

Some species exist in the same environment without directly affecting each other. If they use different resources in the environment and do not prey on each other, their direct interactions are neutral - neither good nor bad for either species. Even if their interactions are neutral, they are part of the same ecosystem, so changes in the ecosystem can affect both species similarly or differently.

Example: [California ground squirrels](#) and [mule deer](#) both live in many California ecosystems. The [squirrels](#) generally eat seeds, barley, oats, and acorns with some insects and bird eggs. In contrast, [mule deer](#) largely browse on bushes and trees, with acorns being an important part of their diet in summer and fall (Sommer et al., 2007). These two species might compete for acorns when other resources are sparse, but they mostly exist in neutral proximity to each other. However, both are hunted by [coyotes](#), [bobcats](#), and [pumas](#), so their populations are linked through other species. For example, if the population of squirrels declines, say from predation by [red tailed hawks](#), there might be more predator pressure on deer from their common predators, e.g. coyotes, bobcats, and pumas. Thus, a large population of squirrels might reduce predation on deer, indirectly benefiting them, while also increasing competition for acorns if other food resources are low.

Competition

Competition for resources and space is very common among species that occupy similar roles in an ecosystem. When species compete with each other, both pay a price for that competition because they are sharing resources. When interactions are predominantly competitive, either species would do better if the other was not present. Competition tends to drive selection of the organisms that are best suited for a particular ecosystem role, with the population of the better adapted species increasing relative to that of the less well adapted species. As the population of the more successful species increases, it is less affected by the competition because each organism experiences fewer interactions with its competitors. In contrast, each individual of the less successful species experiences relatively more competitive interactions, making the harm experiences greater.

Example: California grasslands have been invaded by new species of grass introduced by humans. These new species out competed the native species due to their greater ability to reproduce in many of the environments (see for example, [Corbin and D'Antonio, 2010](#)). The new species are taking up the space and nutrients previously used by the native grasses. In this case, the native species experienced a significantly greater harm during the competition than the invasive species.

Example: [Virginia opossums](#) and [striped skunks](#) actually do compete with each other to some degree. Both are opportunistic feeders, eating almost anything that is available, with opossums eating more small vertebrates and plants and skunks focusing more on insects. They are also about the same size. Thus, they play a similar role in the ecosystem and can compete for both food and dens, although opossums can also den in trees. In contrast to the example of the grasses, opossums and skunks commonly co-exist, with the competition less intense and their abilities to reproduce more equal in many ecosystems.

Commensalism

Sometimes interactions benefit one organism while being of neither benefit nor harm to the other organism. These types of relationships are also very common and help promote diverse ecosystems.

Examples: A [tree frog](#) in a rainforest benefits from the habitat created by the tree it lives in without harming or benefiting the tree. A [cattle egret](#) can catch half again as much food while expending two thirds of the energy if it searches for insects near grazing animals who disturb the insects. This activity does not directly benefit the grazing animals, nor does it harm them.

Amensalism

Interactions can have no effect on one species while harming another. These interactions are usually incidental to a behavior of the species that does not experience harm or benefit. They can include interactions such as a change in the environment, incidental

killing of organisms, and other influences.

Examples: Hippos consume a significant amount of undigestible organic matter, and they create a very large volume of poop. They often poop in ponds, and these ponds can become anoxic, killing the aquatic organisms that require oxygen to live (Pennisi, 2018). Most animals have stepped on insects, killing them.

Parasitism and Predation

Parasitism and predation both have a strong benefit for one species and a significant harm to the other. In both cases, one organism takes resources from another. Parasitism involves one organism living off the resources of another for an extended period of time without causing their death. Parasites benefit from the survival of their host organism. In contrast, predation usually involves killing the prey organism and consuming it. Often, parasites are small relative to their hosts, whereas predators are frequently about the same size as or larger than their prey. This difference in size is related to the energy needs of an organism relative to its size: in general, larger organisms need more energy to accumulate their biomass and to maintain their activity. A small parasite can take energy from a larger host for an extended period of time without killing it. In contrast, a larger organism generally needs more energy, so it is difficult to maintain itself without killing the organism it is preying on. Similarly, it is easier to kill an organism that is of similar size or smaller, so predators tend to prey on smaller organisms. In contrast, parasites typically kill by causing systemic problems for their hosts, such as diseases, extracting too many resources, or taking over cellular processes in the case of viruses.

Parasitism Examples: Parasites include organisms like ticks that live by sucking the blood of mammals. Other parasites include mistletoe living in trees; mistletoe has evolved to plant its "roots" into the circulation system of the trees, extracting nutrients and water from the host. This can cause significant damage to the tree.

Predation Examples: Obvious examples include animals that prey on other animals. (see <https://necsi.edu/predator-prey-relationships> for some examples.)

Extra:

There are a lot of questions as to how herbivory (eating of plants) fits into this relationship scheme. If an herbivore kills the plant, the interaction is similar to predation. However, many plants survive and some have even evolved to do better when fed on by other species. For example, fruits often attract other species to help distribute the seeds for the plant. Thus, the relationships between plants and the things that eat them can be evaluated in terms of harms and benefits much like those involving other species interactions.

Summary

Relationships among organisms can be more complicated than who benefits or loses from the direct interactions. Both direct and indirect interactions have been driving forces for evolution, leading to deeply interconnected communities within ecosystems. The relationships often become more intricate and interdependent through time as species help and harm each other, responding to the organisms around them. These interactions often lead to stabilizing feedbacks and persistent ecosystems. Sometimes, however, a geological event, evolution of a revolutionary process, or the introduction of a new species can disrupt the established relationships and lead to ecological change that triggers an amplifying feedback.

Return to Daisyworld

In the introduction to this chapter, I posed some questions about the interactions among daisies in Daisyworld:

How many different types of interactions are there? Can they be classified into the interaction types described in the second part of the chapter? If this was a real ecosystem, what other organisms would be needed to allow the daisies to grow? How would any of these needs affect the simple Daisyworld model?

How would you answer these questions after reading about different interaction types? What would you like to add to a Daisyworld model?

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<https://animaldiversity.org>

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CHAPTER OVERVIEW

2: What is life?

We only know life from Earth, which makes it difficult to define objectively. As scientists, we use what we know as a model, e.g. life on Earth. These readings help us understand what life is and how we classify it in the context of the evolution of life on Earth - with an understanding that it could be different elsewhere in the Universe.

[1.1: What is life?](#)

Thumbnail: Several types of photosynthetic bacteria (Cyanobacteria) that grow in Antarctic lakes and form pinnacles.

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1.1: What is life?

LEARNING OBJECTIVES

- Understand essential properties of life
- Distinguish between living and non-living systems

If we are going to talk about biology, and organisms and cells and such, we have to define what we mean by life. This raises a problem peculiar to biology as a science. We cannot define life generically because we know of only one type of life. We do not know whether this type of life is the only type of life possible or whether radically different forms of life exist elsewhere in the Universe or even on Earth, in as yet to be recognized forms.

While you might think that we know of many different types of life, from mushrooms to whales, from humans to the bacterial communities growing on the surfaces of our teeth (that is what dental plaque is, after all), the closer we look the more these different “types of life” are in fact all versions of a common underlying motif: they represent versions of a single type of life. Based on their common chemistry, molecular composition, cellular structure, and the way that they encode hereditary information in the form of molecules of deoxyribonucleic acid (DNA), all topics we will consider later on, there is no reasonable doubt that all organisms are related; they are descended from a common ancestor. We will have to deal with viruses, too. Are they life and do they share an common ancestor with the rest of life?

Astrobiology

We cannot currently answer the question of whether the origin of life is a simple, likely, and predictable event given the conditions that existed on the Earth when life first arose, or whether it is an extremely rare and unlikely event. In the absence of empirical data for life elsewhere, it is challenging for astrobiologists to rigorously use scientific methods to search for something - life - that might not exist. However, asking questions that are seemingly impossible to answer, provided that empirically-based interpretations can be generated, has often been a significant driver of scientific progress.

The techniques scientists use to detect and characterize life depend on our current understanding of life. Essentially all of the methods are based on what we already know about life. When we search life in extreme environments on Earth, for example, we often rely on the fact that all known organisms use DNA to encode their genetic information; we use methods to detect and characterize that DNA. However, these methods would not detect organisms that used a non-DNA method to encode genetic information; DNA-based methods would not be expected to recognize dramatically different types of life. Alternatively, we can look for evidence of metabolism - chemical reactions catalyzed by life. However, if metabolism occurs on a different time scale than our observations, we will not recognize it as a product of life.

One challenge with developing alternative methods for detecting dramatically different types of life is that we do not know what to look for. We need models of water alternative life might look like to be able to recognize it. If we could generate living systems de novo in the laboratory or if we better characterize organics on other planets or moons, we would have a better understanding of what functions are necessary for life and how to look for possible “non-standard” organisms using better methods. The research might even lead to the discovery of alternative forms of life on Earth, if they exist.²⁸ We can also work to develop realistic models for alternative forms of life; even though scientists have studied alternative chemical and organic systems, no one has yet been able to construct a model that predicts life-like behavior for any system significantly different from life as we know it on Earth. In the absence of a model for non-standard forms of life, even a theoretical one, the reasonable scientific approach is to concentrate on the characteristics of life as we know them from Earth, while keeping in mind that we may be excluding other types of life in the Universe.

Properties of life

So, let us try to produce a good definition for life, or given the fact that we know only of one version of life, a useful description of what we mean by life. First, the core units of life are organisms, which are individual living objects. From a structural and thermodynamic perspective, each organism is a **bounded, non-equilibrium system that persists over time**. From a practical point of view, an organism needs to be able to produce one or more copies of itself, and it consumes energy to do so. Even though organisms are composed of one or more cells, the organism that is the basic unit of life because it is the organism that reproduces.

Reproduction

Why do we include reproduction in our definition of life? This is basically a pragmatic criterion: organisms have to be able to reproduce to persist through time. Assume that a non-reproducing form of life was possible. A system that could not reproduce runs the risk of death (or perhaps better put, extinction) by accident. Over time, the probability of death for a single individual will approach one – that is, certainty. In contrast, a system that can reproduce makes multiple copies of itself and so minimizes, although by no means eliminates, the chance of accidental extinction, the death of all of its descendants. We see the value of this strategy when we consider the history of life. Even though there have been a number of mass extinction events over the course of life's history,³¹ organisms descended from a common ancestor that appeared billions of years ago, and they continue to survive and flourish - and reproduce.

Metabolism and structure

Metabolism consists of the chemical reactions that occur within organisms that allow them to persist. These reactions are essential to life because they provide energy and raw materials to make the cell structure, control the exchange of materials with the environment, and allow reproduction. Metabolism is the foundation of how life works. Much of an organism's structure is related to facilitating metabolism. Organisms confine their metabolic reactions to within cells, but they also exchange resources with their environment. Organisms are able to import, in a controlled manner, energy and matter from their environment and to export waste products into their environment. To control these fluxes, a distinct boundary between the organism and the rest of the world is required. The basic barrier for all organisms (except viruses - if they are organisms) is a cell membrane, and it appears to be a homologous structure across all organisms – that is, cell membranes were present in and inherited from the common ancestor to all life. The cell membrane modulates the transport of chemicals into and out of cells, and is thus critical to metabolism. The importation of chemicals to produce energy, specifically energy that can be used to drive various cellular processes, is what enables the organism to maintain its non-equilibrium state and its dynamic structure. The boundary with the environment must be able to retain the valuable molecules generated, while at the same time allow waste products to leave. This ability to import matter and export waste enables the organism to grow and to reproduce.

Metabolism allows organisms to persist in disequilibrium with their environment. We see evidence of the non-equilibrium nature of organisms in their activities, including their ability to move, to change their environmental chemistry, and to grow. It is important for all aspects of the living state. In particular, organisms use thermodynamically favorable reaction to capture energy from their environment. They use this energy to drive a wide range of thermodynamically unfavorable chemical reactions, creating disequilibrium with their environment. An organism that reaches thermodynamic equilibrium is a dead organism because it no longer has energy to live.

Non-living structures

It is often useful to think of systems that have some of the properties of life, but are not alive. There are examples of non-living, non-equilibrium systems that can “self-organize” or appear *de novo*. Hurricanes and tornados form spontaneously and then dissipate. They use energy from their environment, which is then dispersed back into the environment, a process associated with increased entropy. These non-living systems differ from organisms in that they cannot produce offspring - they are the result of specific environmental conditions. They are individual entities, unrelated to one another, which do not and cannot evolve through inheritance. Tornados and hurricanes that formed billions or millions of years ago would (if we could observe them) be similar to those that form today because they emerge from the same processes. Since we understand (more or less) the conditions that produce tornados and hurricanes, we can predict, fairly reliably, the conditions that will lead to their appearance and how they will behave once they form. In contrast, organisms that lived in the past were different from those that are alive today. The further into the past we go, the more different they appear. Some ancient organisms became extinct, some gave rise to the ancestors of current organisms. In contrast, modern tornados and hurricanes originate anew, they are not derived from parental storms.

Question to ponder:

- Speculate on how you might look for evidence of life on another planet or moon. What are characteristics might be universal to all life and what might be specific to life on Earth?
- Are viruses alive? You might not know much about them now, so feel free to speculate. How would you look for evidence of something like a virus on another planet or moon?
- Make a model of what properties a biological boundary with its environment needs to possess. Using your current knowledge, how would you build such a boundary layer?
- Ponder the non-equilibrium nature of a specific form of life, maybe your pet or something from your favorite natural area. What about it tells you it is alive? What happens to it when it dies? How does that relate to thermodynamics and equilibrium?
- Ponder a non-living structure that has a pattern. How can you tell that it is not living? For example, how might you be able to demonstrate that the great red spot on Jupiter is not a life form?
- What do you think about considering Earth and all the life on it as a single organism, similar to the Gaia concept? What aspects of the Earth-life system fit a traditional definition of life and which do not?

Additional Reading

- How astrobiologist Chris McKay thinks about "What is life?"
- Dozens of answers to the Question of the Month "What is life?" on Philosophy Now

References

²⁸ The possibility of alternative microbial life on Earth: <http://www.ncbi.nlm.nih.gov/pubmed/18053938> Signatures of a shadow biosphere: <http://www.ncbi.nlm.nih.gov/pubmed/19292603>; Life on Earth but not as we know it (with the caveat that this is a relatively old article; we now know that most of life in desert varnish is related to the rest of know life; however, this article is still an interesting example of the arguments that scientists have on what is actually present on Earth): <https://www.theguardian.com/science/2013/apr/14/shadow-biosphere-alien-life-on-earth>

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CHAPTER OVERVIEW

3: Genes to Enzymes

DNA, RNA, Genes and Chromosomes

DNA Replication

How Asexual Prokaryotes Achieve Genetic Diversity

Mutations

Operon Theory

Protein Synthesis (Translation)

RNA Transcription

What Are Genes?

Thumbnail: DNA Double Helix. Image used with permission (Public Domain; [ApersOn](#)).

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DNA, RNA, Genes and Chromosomes

skills to develop

- Explain the differences between RNA and DNA in terms of function
- Recognize the structure of DNA
- Explain the two functions of the genome
- Explain how DNA and genes are converted into proteins (the "central dogma")
- Differentiate between genotype and phenotype and explain how environmental factors influence phenotype
- Describe why DNA is packaged into chromosomes

Nucleic Acids

Nucleic acids are macromolecules that all life on Earth uses to hold its evolutionary memory. They carry the genetic blueprint of a cell in the form of deoxyribonucleic acid (DNA) and the instructions for the function of the cell in the form of ribonucleic acid (RNA). Chromosomes are composed of DNA arranged into genes and other non-coding areas. The genes are the blueprints for making various molecules that cells need to function, including enzymes. The non-coding areas are critical for influencing when genes are activated or expressed and play diverse other roles in cell function, some of which are not yet understood. DNA is a relatively stable molecule that is tightly controlled by the host cell. RNA is much more reactive than DNA. It plays diverse, reactive functions in cells. When most genes are expressed, they are translated into messenger RNA (mRNA) molecules, which are then transcribed into proteins. Some genes produce RNA molecules that play critical functions in cells, such as transcribing mRNA into proteins (ribosomal RNA or rRNA), transporting amino acids (transfer RNA or tRNA), and influencing gene expression (microRNA). RNA is so fundamental to basic cell functions that some scientists think that life originated in a "RNA world" with the transition from abiotic organic chemistry to life requiring the catalytic properties of RNA molecules (See <https://www.khanacademy.org/science/ap-biology/natural-selection/origins-of-life-on-earth/a/rna-world> for more information).

Although RNA is critical to cell function, it is a reactive molecule and does not provide a reliable memory system for inheritance in organisms. In contrast, the structure of DNA molecules makes them much less reactive, and they can be packaged into nested spirals that protect them from damage. Thus, DNA is a more reliable memory system for inheritance. It is used by all organisms (except some viruses – if they are organisms) to encode critical cell functions. Most of the DNA in cells is packaged into chromosomes that are copied when cells reproduce. However, some DNA is present in plasmids (small circular DNA molecules outside of chromosomes) and Eukaryotic organelles (e.g. plastids and mitochondria). The entire genetic content of a cell is known as its genome.

What are DNA and RNA?

Nucleic acids are made up of nucleotides composed of a nitrogenous base, a pentose sugar, and a phosphate group. The nitrogenous bases have 5 different structures and are the letters that compose the codes in DNA and RNA molecules. These bases include: adenine (A), guanine (G) cytosine (C), thymine (T) in DNA, and uracil (U) in RNA (figure showing the structure of the bases). DNA and RNA molecules consist of chains of nitrogenous bases bound together by their attached phosphate groups.

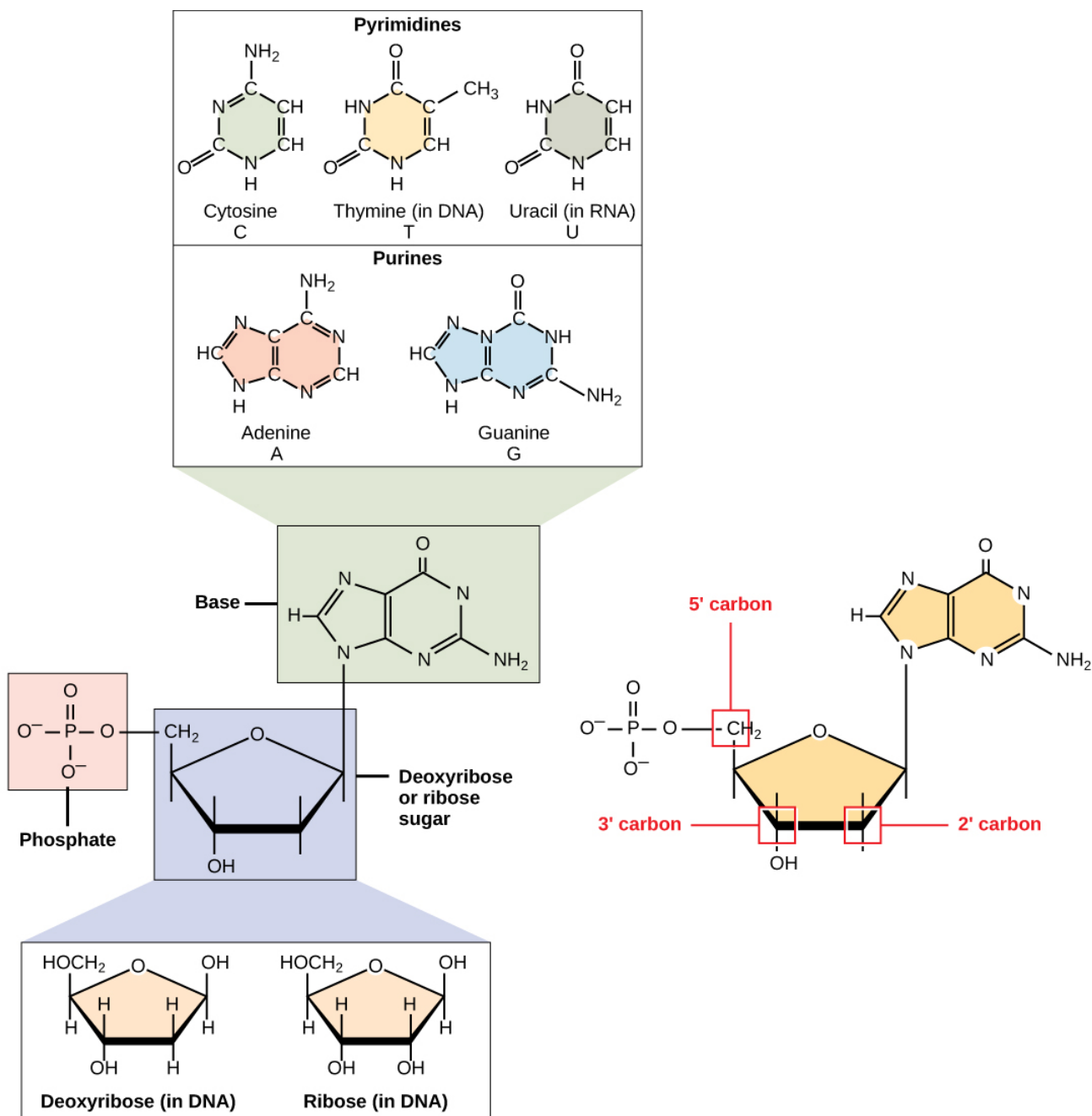


Figure Nucleotides: A nucleotide is made up of three components: a nitrogenous base, a pentose sugar, and one or more phosphate groups. Carbon residues in the pentose are numbered 1' through 5' (the prime distinguishes these residues from those in the base, which are numbered without using a prime ' notation). The base is attached to the 1' position of the ribose, and the phosphate is attached to the 5' position. When a nucleotide is added to a DNA or RNA molecule, the 5' phosphate of the incoming nucleotide attaches to the 3' hydroxyl group at the end of the growing chain. Two types of pentose are found in nucleotides, deoxyribose (found in DNA) and ribose (found in RNA). Deoxyribose is similar in structure to ribose, but it has an H instead of an OH at the 2' position. Bases can be divided into two categories: purines and pyrimidines. Purines have a double ring structure, and pyrimidines have a single ring. Variations in the structures of these molecules affect their function, differentiating DNA from RNA and the reactivity of the molecules.

In DNA, two strands of nucleotides are paired together by hydrogen bonds between their bases in a very systematic way. A can bond with T, and G can bond with C when the two strands are oriented in opposite directions. The two strands only pair if their sequences of bases are complementary. For example, if the sequence of one strand is AATTGGCC, the complementary strand

would have the sequence TTAACCGG. When each base is matched to its complement on the other strand, the two strands spiral around each other, creating a very stable (and famous) double-helical structure.

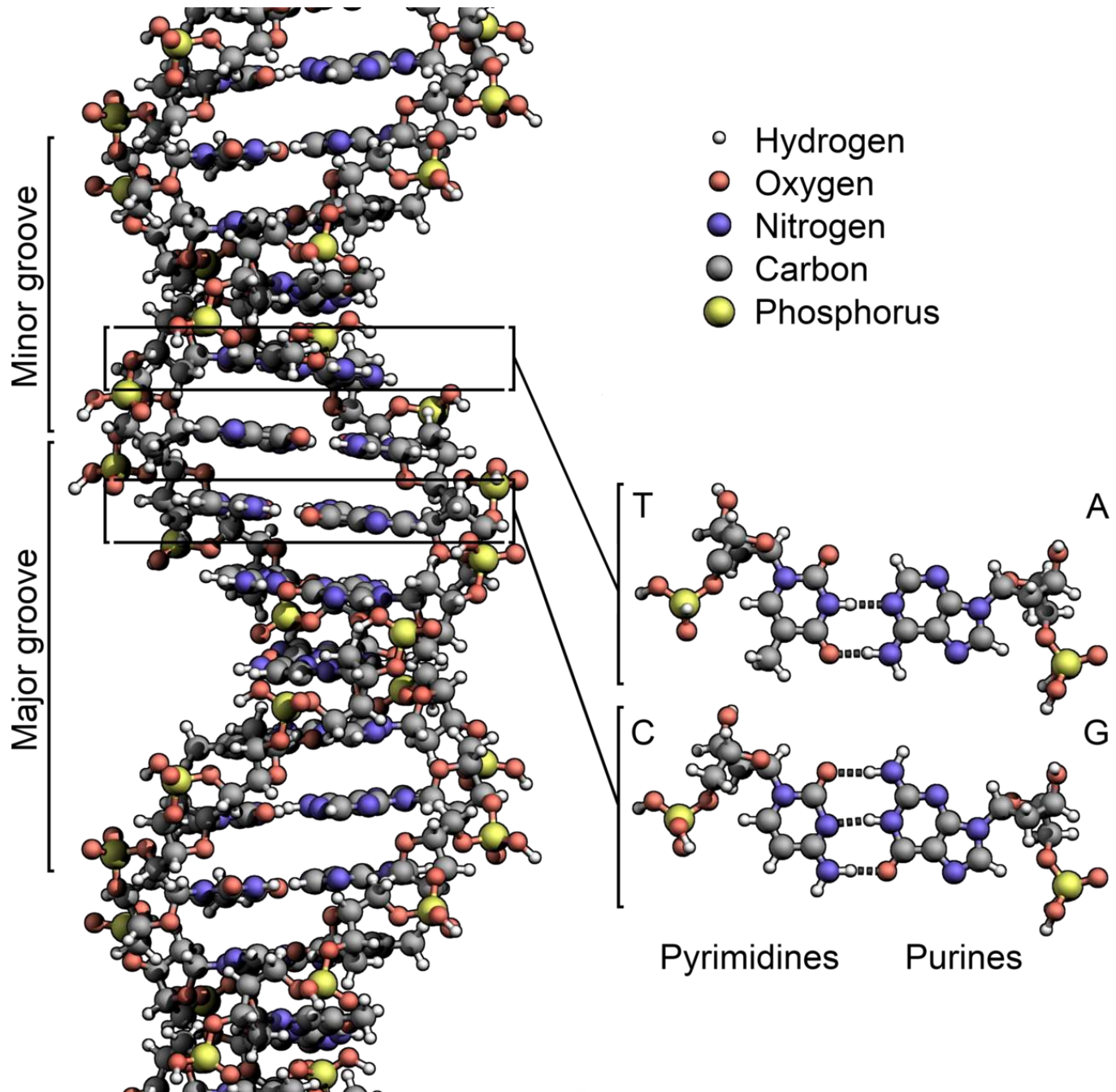


Figure DNA: The structure of DNA showing the details of hydrogen bonds (dashed lines) between complementary base pairs. Note that the yellow phosphate groups point upward on the right side and downward on the left side. The grooves are due to asymmetry in the spiral. (Figure credit: Zephyris (Richard Wheeler))

In contrast to DNA, RNA is single-stranded, and the nitrogenous bases are not necessarily paired with their complements. (U takes the place of T in RNA, bonding with A.) The bases are, however, reactive. If there are stretches of complementary base pairs elsewhere on the RNA molecule or on another RNA molecule, those segments will form hydrogen bonds, tying the two complementary parts together. When sections of RNA bind with each other, the RNA strand(s) folds into specific shapes, which affect how they catalyze reactions. The patterns of RNA base pairs determine how the strands fold, allowing them to perform different functions for the cells.

Management of DNA and RNA is critical for organisms to live and reproduce.

Genes

DNA is the genetic material responsible for inheritance and is passed from parent to offspring for all life on Earth. To preserve the integrity of this genetic information, DNA must be replicated with great accuracy, with minimal errors that introduce changes to the DNA sequence. A genome contains the full complement of DNA within a cell and is organized into smaller, discrete units called genes that are arranged on chromosomes and plasmids.

A gene is composed of DNA that is “read” or transcribed to produce an RNA molecule during the process of transcription. One major type of RNA molecule, called messenger RNA (mRNA), provides the information for the ribosome to catalyze protein synthesis in a process called translation. The processes of transcription and translation are collectively referred to as gene expression. Gene expression is the synthesis of a specific protein with a sequence of amino acids that is encoded in the gene. The flow of genetic information from DNA to RNA to protein is described by the central dogma (Figure 11.1.111.1.1). Each of the processes of replication, transcription, and translation includes the stages of 1) initiation, 2) elongation (polymerization), and 3) termination. These stages will be described in more detail in this chapter.



Figure 11.1.111.1.1: The central dogma states that DNA encodes messenger RNA, which, in turn, encodes protein.

A cell's genotype is the full collection of genes it contains, whereas its phenotype is the set of observable characteristics that result from those genes. The phenotype is the product of the array of proteins being produced by the cell at a given time, which is influenced by the cell's genotype as well as interactions with the cell's environment. Genes code for proteins that have functions in the cell. Although a cell's genotype remains constant, not all genes are used to direct the production of their proteins simultaneously. Cells carefully regulate expression of their genes, only using genes to make specific proteins when those proteins are needed (Figure 11.1.211.1.2). Thus, their phenotype changes through time.

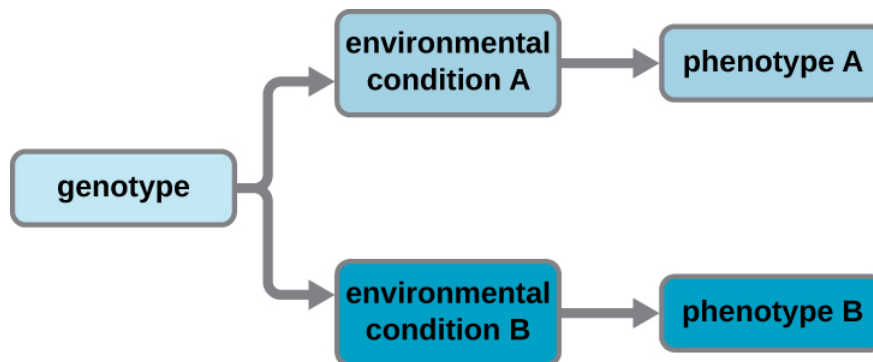


Figure 11.1.211.1.2: Phenotype is determined by the specific genes within a genotype that are expressed under specific conditions. Although multiple cells may have the same genotype, they may exhibit a wide range of phenotypes resulting from differences in patterns of gene expression in response to different environmental conditions.

Chromosomes

DNA strands are packaged into chromosomes with proteins, which allow cells to control gene expression (and phenotypes), replication of the DNA, and separation of the genetic material into cells as they divide. Most (but not all) Bacteria and Archaea have circular chromosomes, with the DNA strands forming a continuous loop. These loops are often twisted up on themselves so the chromosomes make compact structures in the cells (<https://www.nature.com/scitable/topicpage/genome-packaging-in-prokaryotes-the-circular-chromosome-9113/>). The region in a cell with the condensed chromosome is called the nucleoid. Some DNA can also be housed in structures called plasmids that are more easily exchanged between organisms.

Eukaryotes organize their DNA differently. First, most of the DNA is encased in the cell nucleus, which is separated from the rest of the cell with a membrane. Within the nucleus, the DNA strands are packed into multiple linear chromosomes. DNA strands are wrapped around special proteins called histones (also present in some Archaea) and then folded into even tighter areas by additional proteins. The folding of the chromosomes and the details of the chemical structure of long chains coming off the histones strongly influence whether or not genes on the chromosomes can be expressed. Eukaryotes also have DNA in other parts of their cells, specifically certain organelles, including mitochondria and plastids (which include chloroplasts). The DNA in

organelles (orDNA) is inherited from ancient bacteria that were the ancestors of these organelles. orDNA is packaged into their own chromosomes, whose structure varies among different organisms and even during development of a single organism. In general, orDNA structure is highly variable and poorly understood relative to nuclear DNA structure and replication (see <https://www.frontiersin.org/articles/10.3389/fpls.2015.00883/full>)

For all DNA, the structure of the chromosomes strongly influences how and when the DNA is active and can be copied.

Key Concepts and Summary

- Nucleic acids are molecules composed of three components. The details of these components affect the function and behavior of RNA and DNA molecules
- The sequence of nitrogenous bases in DNA and RNA provides the information recorded in the molecules.
- Double-stranded DNA has a helical structure in which bases are paired with their complements on the opposite strand.
- Genes consist of DNA sequences that code for the production of specific proteins.
- The genotype of a cell consists of all of its genes and remains constant. The phenotype of a cell consists of the functions a cell is capable of due to the expression of genes. Phenotype varies through time as gene expression changes in response to environmental and biochemical signals.
- Organisms organize their DNA into chromosomes. The structure of the chromosomes changes through time and affects which genes can be expressed.

More Information

RNA World: <https://www.khanacademy.org/science/ap-biology/natural-selection/origins-of-life-on-earth/a/rna-world>

Structure of bacterial chromosomes: <https://www.nature.com/scitable/topicpage/genome-packaging-in-prokaryotes-the-circular-chromosome-9113/>

Organelle DNA organization: <https://www.frontiersin.org/articles/10.3389/fpls.2015.00883/full>

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